第九章 正则变换 哈密顿-雅可比方程

引：哈密顿方程对正则变换具有不变性

§9.1 正则变换

1. 正则变换的条件

Lagrange力学：

目标：通过点变换（广义坐标的变换），增加可遗坐标个数

Hamilton力学： q_α, p_α 等价（正则性）

考虑变换 $\begin{cases} P_\alpha = P_\alpha(p,q,t) \\ Q_\alpha = Q_\alpha(p,q,t) \end{cases}$ 正则变换：P_α, Q_α 仍满足Hamilton方程 $\begin{cases} \dot{Q}_i = \dfrac{\partial H'}{\partial P_i} \\ \dot{P}_i = -\dfrac{\partial H'}{\partial Q_i} \end{cases}$

P_α、Q_α：一对正则变量 / 共轭变量

哈密顿方程成立 ⟺ 最小作用量原理

$$\delta \int_{t_1}^{t_2} \left[\sum_\alpha p_\alpha \dot{q}_\alpha - H(p,q,t)\right] dt = 0$$

$$\Updownarrow$$

$$\delta \int_{t_1}^{t_2} \left[\sum_\alpha P_\alpha \dot{Q}_\alpha - K(P,Q,t)\right] dt = 0$$

其中K是变换后的哈密顿函数

$$\Rightarrow \left[\sum_\alpha p_\alpha \dot{q}_\alpha - H(p,q,t)\right] - \left[\sum_\alpha P_\alpha \dot{Q}_\alpha - K(P,Q,t)\right] = \frac{dU}{dt}$$

即 $\sum_\alpha p_\alpha dq_\alpha - \sum P_\alpha dQ_\alpha + (K-H)dt = dU$

大辛条件：

$$Q = Q(q,p), \quad P = P(q,p), \quad K = K(Q,P)$$

$$\dot{Q} = \frac{\partial Q}{\partial q}\dot{q} + \frac{\partial Q}{\partial p}\dot{p} = \frac{\partial Q}{\partial q}\frac{\partial H}{\partial p} - \frac{\partial Q}{\partial p}\frac{\partial H}{\partial q}$$

$$\dot{P} = \frac{\partial P}{\partial q}\dot{q} + \frac{\partial P}{\partial p}\dot{p} = \frac{\partial P}{\partial q}\frac{\partial H}{\partial p} - \frac{\partial P}{\partial p}\frac{\partial H}{\partial q}$$

H、K 不含时，$H = K$

$$\frac{\partial H}{\partial q} = \frac{\partial K}{\partial q} = \frac{\partial K}{\partial Q}\frac{\partial Q}{\partial q} + \frac{\partial K}{\partial P}\frac{\partial P}{\partial q}, \quad \frac{\partial H}{\partial p} = \frac{\partial K}{\partial p} = \frac{\partial K}{\partial Q}\frac{\partial Q}{\partial p} + \frac{\partial K}{\partial P}\frac{\partial P}{\partial p}$$

$$\begin{pmatrix}\dot{Q}\\ \dot{P}\end{pmatrix} = \begin{pmatrix}\frac{\partial Q}{\partial q}\frac{\partial H}{\partial p} - \frac{\partial Q}{\partial p}\frac{\partial H}{\partial q}\\ \frac{\partial P}{\partial q}\frac{\partial H}{\partial p} - \frac{\partial P}{\partial p}\frac{\partial H}{\partial q}\end{pmatrix} = \begin{pmatrix}\frac{\partial Q}{\partial q} & \frac{\partial Q}{\partial p}\\ \frac{\partial P}{\partial q} & \frac{\partial P}{\partial p}\end{pmatrix}\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}\begin{pmatrix}\frac{\partial H}{\partial q}\\ \frac{\partial H}{\partial p}\end{pmatrix}$$

$$= \begin{pmatrix}\frac{\partial Q}{\partial q} & \frac{\partial Q}{\partial p}\\ \frac{\partial P}{\partial q} & \frac{\partial P}{\partial p}\end{pmatrix}\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}\begin{pmatrix}\frac{\partial Q}{\partial q} & \frac{\partial P}{\partial q}\\ \frac{\partial Q}{\partial p} & \frac{\partial P}{\partial p}\end{pmatrix}\begin{pmatrix}\frac{\partial K}{\partial Q}\\ \frac{\partial K}{\partial P}\end{pmatrix}$$

$$= J W J^T \begin{pmatrix}\frac{\partial K}{\partial Q}\\ \frac{\partial K}{\partial P}\end{pmatrix}$$

$$= W \begin{pmatrix}\frac{\partial K}{\partial Q}\\ \frac{\partial K}{\partial P}\end{pmatrix}$$ 哈密顿方程仍成立.

$\Rightarrow \quad J W J^T = W$

2. 母函数

称函数 U_1 为正则变换的母函数

$U = U_1(\underline{q}, \underline{Q}, \underline{t})$ 选取 q, Q 为自变量

$$\Rightarrow \begin{cases} p_\alpha = \frac{\partial U_1}{\partial q_\alpha}, \quad P_\alpha = -\frac{\partial U_1}{\partial Q_\alpha} \\ K - H = \frac{\partial U}{\partial t} \end{cases}$$

Legendre 变换

$U_2(p, Q, t) = U_1(q, Q, t) - \sum_\alpha p_\alpha q_\alpha$ 以 p、Q 为自变量

$dU_2 = dU_1 - \sum_\alpha p_\alpha dq_\alpha - \sum_\alpha q_\alpha dp_\alpha$

$\quad = -\sum_\alpha q_\alpha dp_\alpha - \sum_\alpha P_\alpha dQ_\alpha + (K-H)dt$

$$\Rightarrow \begin{cases} q_\alpha = -\dfrac{\partial U_2}{\partial p_\alpha}, \quad P_\alpha = -\dfrac{\partial U_2}{\partial Q_\alpha} \\ K - H = \dfrac{\partial U_2}{\partial t} \end{cases}$$

$U_3(q, P, t) = U_1(q, Q, t) + \sum_\alpha P_\alpha Q_\alpha$ 以 q, P 为自变量

$$dU_3 = dU_1 + \sum_\alpha P_\alpha \, dQ_\alpha + \sum_\alpha Q_\alpha \, dP_\alpha$$

$$= \sum_\alpha p_\alpha \, dq_\alpha + \sum_\alpha Q_\alpha \, dP_\alpha + (K - H)\, dt$$

$$\Rightarrow \left\{ p_\alpha = \frac{\partial U_3}{\partial q_\alpha} \quad Q_\alpha = \frac{\partial U_3}{\partial P_\alpha} \right.$$

$U_4(p, P, t) = U_1(p, q, t) - \sum_\alpha p_\alpha q_\alpha + \sum_\alpha P_\alpha Q_\alpha$ 以 p, P 为自变量

$$dU_4 = -\sum_\alpha q_\alpha \, dp_\alpha + \sum_\alpha Q_\alpha \, dP_\alpha + (K - H)\, dt$$

$$\Rightarrow \begin{cases} q_\alpha = -\dfrac{\partial U_4}{\partial p_\alpha}, \quad Q_\alpha = \dfrac{\partial U_4}{\partial P_\alpha} \\ K - H = \dfrac{\partial U_4}{\partial t} \end{cases}$$

3. 正则变换举例

例1：恒等变换

取 $U = U_3(q, P, t) = \sum_\alpha q_\alpha P_\alpha$

$$\Rightarrow p_\alpha = \frac{\partial U_3}{\partial q_\alpha} = P_\alpha, \quad Q_\alpha = \frac{\partial U_3}{\partial P_\alpha} = q_\alpha$$

例2：量纲/伸缩变换

$$U_3(q, P, t) = \sum_\alpha k_\alpha q_\alpha P_\alpha$$

$$\Rightarrow p_\alpha = \frac{\partial U_3}{\partial q_\alpha} = k_\alpha P_\alpha \quad Q_\alpha = \frac{\partial U_3}{\partial P_\alpha} = k_\alpha q_\alpha$$

即 $P_\alpha = \dfrac{p_\alpha}{k_\alpha}$　　原因：作用量的量纲 ML^2T^{-1} 不变

$Q_\alpha = k_\alpha q_\alpha$　　/ $p = \dfrac{\partial L}{\partial \dot{q}}$

例3：点变换（仅为坐标的变换）

$$U_3(q, P, t) = \sum_\alpha f_\alpha(q, t) P_\alpha$$

$$\Rightarrow \left\{ p_\alpha = \frac{\partial U_3}{\partial q_\alpha} = \sum_\beta \frac{\partial f_\beta}{\partial q_\alpha} P_\beta \right.$$

$$Q_\alpha = \frac{\partial u_3}{\partial P_\alpha} = f_\alpha(q,t)$$

* 当 $f_\alpha = \sum_\beta a_{\alpha\beta} q_\beta$ 且满足正交条件：

$$\sum_\beta a_{\alpha\beta} a_{\gamma\beta} = \delta_{\alpha\gamma}$$

称 $Q_\alpha = \sum_\beta a_{\alpha\beta} q_\beta$ 为正交变换

而 $p_\alpha = \sum_\beta a_{\beta\alpha} P_\beta$

) 同乘 $a_{\gamma\alpha}$，对 α 求和

$$\sum_\alpha a_{\gamma\alpha} p_\alpha = \sum_\alpha \sum_\beta a_{\gamma\alpha} a_{\beta\alpha} P_\beta = \sum_\beta \delta_{\beta\gamma} P_\beta = P_\gamma$$

偷偷交换了求和顺序.

例4：

$$u_1(q,Q,t) = \sum_\alpha q_\alpha Q_\alpha$$

$$p_\alpha = \frac{\partial u_1}{\partial q_\alpha} = Q_\alpha, \quad P_\alpha = -\frac{\partial u_1}{\partial Q_\alpha} = -q_\alpha$$

$\Rightarrow$ P、q 是对等的

例5：利用正则变换解决谐振子问题.

$$H = \frac{P^2}{2m} + \frac{1}{2}kx^2$$

解：取 $U_1(x,X,t) = \frac{1}{2}\sqrt{km}\,x^2 \cot(2\pi X)$

$$\begin{cases} p = \frac{\partial u_1}{\partial x} = \sqrt{km}\,x \cot(2\pi X) \\ P = -\frac{\partial u_1}{\partial X} = \pi\sqrt{km}\,x^2 \csc^2(2\pi X) \end{cases}$$

解得 $\begin{cases} x = \sqrt{\frac{P}{\pi\sqrt{km}}} \cdot \sin(2\pi X) \\ p = \sqrt{\frac{P}{\pi}\sqrt{km}} \cdot \cos(2\pi X) \end{cases}$

$$K(P,X) = H(p,x) = \frac{p^2}{2m} + \frac{1}{2}kx^2$$

$$= \frac{P}{2\pi}\sqrt{\frac{k}{m}}$$

X 是循环坐标.

$$\dot{P} = -\frac{\partial K}{\partial X} = 0 \qquad \dot{X} = \frac{\partial K}{\partial P} = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

可设 $X = \theta + \frac{1}{2\pi}\sqrt{\frac{k}{m}}\,t$

解得

$$x = \sqrt{\frac{P}{\pi\sqrt{km}}}\,\sin\left(\sqrt{\frac{k}{m}}\,t + 2\pi\theta\right)$$

4. Poisson 括号的不变性

泊松括号在正则变换下不变 $[\varphi,\psi]_{pq} = [\varphi,\psi]_{PQ}$

$$[\varphi,\psi]_{pq} = \sum_\alpha\left(\frac{\partial\varphi}{\partial q_\alpha}\frac{\partial\psi}{\partial p_\alpha} - \frac{\partial\varphi}{\partial p_\alpha}\frac{\partial\psi}{\partial q_\alpha}\right)$$

$f = f(q, p(q,P), t)$ $\quad p = p(q,P) \quad Q = Q(q,P)$

$\quad = f(Q(q,P), P, t)$

则

$$\frac{\partial f}{\partial q_\alpha} = \frac{\partial}{\partial q_\alpha} + \sum_\beta \frac{\partial P_\beta}{\partial q_\alpha}\frac{\partial}{\partial P_\beta} = \sum_\beta \frac{\partial Q_\beta}{\partial q_\alpha}\frac{\partial}{\partial Q_\beta}$$

$$\Rightarrow \frac{\partial}{\partial q_\alpha} = \sum_\beta\left(\frac{\partial Q_\beta}{\partial q_\alpha}\frac{\partial}{\partial Q_\beta} - \frac{\partial P_\beta}{\partial q_\alpha}\frac{\partial}{\partial P_\beta}\right),$$

同理：
$$\frac{\partial}{\partial p_\alpha} = \sum_\beta\left(\frac{\partial P_\beta}{\partial p_\alpha}\frac{\partial}{\partial P_\beta} - \frac{\partial Q_\beta}{\partial p_\alpha}\frac{\partial}{\partial Q_\beta}\right)$$

$$[\varphi,\psi]_{pq} = \sum_\alpha\left[\sum_\beta\left(\frac{\partial Q_\beta}{\partial q_\alpha}\frac{\partial\varphi}{\partial Q_\beta} - \frac{\partial P_\beta}{\partial q_\alpha}\frac{\partial\varphi}{\partial P_\beta}\right)\frac{\partial\psi}{\partial p_\alpha} - \sum_\beta\left(\frac{\partial Q_\beta}{\partial q_\alpha}\frac{\partial\psi}{\partial Q_\beta} - \frac{\partial P_\beta}{\partial q_\alpha}\frac{\partial\psi}{\partial P_\beta}\right)\frac{\partial\varphi}{\partial p_\alpha}\right]$$

$$= \sum_\alpha\sum_\beta\left(\frac{\partial Q_\beta}{\partial q_\alpha}\frac{\partial\varphi}{\partial Q_\beta}\frac{\partial\psi}{\partial p_\alpha} - \frac{\partial P_\beta}{\partial q_\alpha}\frac{\partial\varphi}{\partial P_\beta}\frac{\partial\psi}{\partial p_\alpha} - \frac{\partial Q_\beta}{\partial q_\alpha}\frac{\partial\psi}{\partial Q_\beta}\frac{\partial\varphi}{\partial p_\alpha} + \frac{\partial P_\beta}{\partial q_\alpha}\frac{\partial\psi}{\partial P_\beta}\frac{\partial\varphi}{\partial p_\alpha}\right)$$

$\frac{\partial u}{\partial P_\alpha} = Q_\alpha$
$$= \sum_\alpha\sum_\beta\frac{\partial Q_\beta}{\partial q_\alpha}\left(\frac{\partial\varphi}{\partial Q_\beta}\frac{\partial\psi}{\partial p_\alpha} - \frac{\partial\psi}{\partial Q_\beta}\frac{\partial\varphi}{\partial p_\alpha}\right) + \frac{\partial P_\beta}{\partial q_\alpha}\left(\frac{\partial\psi}{\partial P_\beta}\frac{\partial\varphi}{\partial p_\alpha} - \frac{\partial\psi}{\partial p_\alpha}\frac{\partial\varphi}{\partial P_\beta}\right)$$

$\frac{\partial u}{\partial q_\alpha} = p_\alpha$
$$= \sum_\alpha\sum_\beta\frac{\partial^2 u}{\partial P_\beta\partial q_\alpha}\left(\frac{\partial\varphi}{\partial Q_\beta}\frac{\partial\psi}{\partial p_\alpha} - \frac{\partial\psi}{\partial Q_\beta}\frac{\partial\varphi}{\partial p_\alpha}\right) + \underbrace{\frac{\partial^2 u}{\partial q_\beta\partial q_\alpha}\left(\frac{\partial\psi}{\partial p_\beta}\frac{\partial\varphi}{\partial p_\alpha} - \frac{\partial\psi}{\partial p_\alpha}\frac{\partial\varphi}{\partial p_\beta}\right)}_{\alpha\leftrightarrow\beta\ \text{变为负}\ \Rightarrow\ =0}$$

$$= \sum_\alpha\sum_\beta\frac{\partial^2 u}{\partial P_\beta\partial q_\alpha}\left(\frac{\partial\varphi}{\partial Q_\beta}\frac{\partial\psi}{\partial p_\alpha} - \frac{\partial\psi}{\partial Q_\beta}\frac{\partial\varphi}{\partial p_\alpha}\right) \quad ①$$

$$[\varphi,\psi]_{PQ} = \sum_\alpha\left(\frac{\partial\varphi}{\partial Q_\alpha}\frac{\partial\psi}{\partial P_\alpha} - \frac{\partial\varphi}{\partial P_\alpha}\frac{\partial\psi}{\partial Q_\alpha}\right)$$

$$= \sum_\alpha\sum_\beta\left(\frac{\partial\varphi}{\partial Q_\alpha}\frac{\partial\psi}{\partial p_\beta}\frac{\partial p_\beta}{\partial P_\alpha} - \frac{\partial\varphi}{\partial Q_\alpha}\frac{\partial\psi}{\partial Q_\beta}\frac{\partial Q_\beta}{\partial P_\alpha} - \frac{\partial\psi}{\partial Q_\alpha}\frac{\partial\varphi}{\partial p_\beta}\frac{\partial p_\beta}{\partial P_\alpha} + \frac{\partial\psi}{\partial Q_\alpha}\frac{\partial\varphi}{\partial Q_\beta}\frac{\partial Q_\beta}{\partial P_\alpha}\right)$$

$\frac{\partial u}{\partial P_\alpha} = Q_\alpha$
$$= \sum_\alpha\sum_\beta\frac{\partial p_\beta}{\partial P_\alpha}\left(\frac{\partial\varphi}{\partial Q_\alpha}\frac{\partial\psi}{\partial p_\beta} - \frac{\partial\psi}{\partial Q_\alpha}\frac{\partial\varphi}{\partial p_\beta}\right) + \frac{\partial Q_\beta}{\partial P_\alpha}\left(\frac{\partial\psi}{\partial Q_\alpha}\frac{\partial\varphi}{\partial Q_\beta} - \frac{\partial\varphi}{\partial Q_\alpha}\frac{\partial\psi}{\partial Q_\beta}\right)$$

$\frac{\partial u}{\partial q_\alpha} = p_\alpha$
$$= \sum_\alpha\sum_\beta\frac{\partial^2 u}{\partial q_\beta\partial P_\alpha}\left(\frac{\partial\varphi}{\partial Q_\alpha}\frac{\partial\psi}{\partial p_\beta} - \frac{\partial\psi}{\partial Q_\alpha}\frac{\partial\varphi}{\partial p_\beta}\right) + \underbrace{\frac{\partial^2 u}{\partial P_\beta\partial P_\alpha}\left(\frac{\partial\psi}{\partial Q_\alpha}\frac{\partial\varphi}{\partial Q_\beta} - \frac{\partial\varphi}{\partial Q_\alpha}\frac{\partial\psi}{\partial Q_\beta}\right)}_{=0}$$

$$= \sum_\alpha\sum_\beta\frac{\partial^2 u}{\partial q_\beta\partial P_\alpha}\left(\frac{\partial\varphi}{\partial Q_\alpha}\frac{\partial\psi}{\partial p_\beta} - \frac{\partial\psi}{\partial Q_\alpha}\frac{\partial\varphi}{\partial p_\beta}\right) \quad ②$$

$\alpha\leftrightarrow\beta$, ① = ②

$\Rightarrow$ $[\varphi,\psi]_{pq} = [\varphi,\psi]_{PQ}$ 泊松括号具有正则不变性.

5. 无限小正则变换

选取母函数为 $U_3 = \sum_\alpha q_\alpha P_\alpha + \varepsilon G(q,P)$

$$\rightarrow \begin{cases} p_\alpha = \frac{\partial U_3}{\partial q_\alpha} = P_\alpha + \varepsilon \frac{\partial G}{\partial q_\alpha} \\ Q_\alpha = \frac{\partial U_3}{\partial P_\alpha} = q_\alpha + \varepsilon \frac{\partial G}{\partial P_\alpha} \end{cases}$$

$$\hookrightarrow \begin{cases} P_\alpha = p_\alpha - \varepsilon \frac{\partial G}{\partial q_\alpha} \\ Q_\alpha = q_\alpha + \varepsilon \frac{\partial G}{\partial P_\alpha} \approx q_\alpha + \varepsilon \frac{\partial G}{\partial p_\alpha} \end{cases}$$

$U_3 = \sum_\alpha q_\alpha P_\alpha + \varepsilon G(q,P)$

$$\begin{cases} dq_\alpha = Q_\alpha - q_\alpha = \varepsilon \frac{\partial G}{\partial p_\alpha} \\ dp_\alpha = P_\alpha - p_\alpha = -\varepsilon \frac{\partial G}{\partial q_\alpha} \end{cases}$$

1. 选取 $G(p,q) = p_\beta$

$$\begin{cases} dq_\alpha = \delta_{\alpha\beta}\,\varepsilon \\ dp_\alpha = 0 \end{cases}$$

(q_β 在相空间中移动 ε)

2. 选取 $G = H(q,p,t)$ ， $dt = \varepsilon$

$$\begin{cases} dq_\alpha = dt \frac{\partial H}{\partial p_\alpha} \\ dp_\alpha = -dt \frac{\partial H}{\partial q_\alpha} \end{cases} \rightarrow \begin{cases} \dot{q}_\alpha = \frac{\partial H}{\partial p_\alpha} \\ \dot{p}_\alpha = -\frac{\partial H}{\partial q_\alpha} \end{cases}$$

$\Rightarrow$ 力学系统的演变，对应一个正则变换在时间中的连续伸展.

对于不显含时的物理量 u

$$du = \sum_\alpha \frac{\partial u}{\partial q_\alpha} dq_\alpha + \frac{\partial u}{\partial p_\alpha} dp_\alpha$$

$$= \varepsilon \sum_\alpha \left(\frac{\partial u}{\partial q_\alpha} \frac{\partial G}{\partial p_\alpha} - \frac{\partial u}{\partial p_\alpha} \frac{\partial G}{\partial q_\alpha} \right)$$

$$= \varepsilon\, [u, G]_{q,p}$$

$u = H$

$$dH = \varepsilon\, [H, G]_{q,p}$$

假如 G 是不显含时的运动积分 ($\frac{dG}{dt}$, $\frac{\partial G}{\partial t} = 0$) 则 $[H, G] = -\frac{\partial G}{\partial t} = 0$

$\hookrightarrow$ $dH = 0$

以 G 为母函数的无穷小变换保持 H 不变

二) 角量运动积分

例：系统 $H = \frac{1}{2m}(p_x^2 + p_y^2 + p_z^2) + V(r)$。

$r = \sqrt{x^2 + y^2 + z^2}$

取 $G = L_z = (\vec{r} \times \vec{p})_z = x p_y - y p_x$ 为生成函数

$[H, L_z] = \frac{1}{2m}[(p_x^2 + p_y^2 + p_z^2), (x p_y - y p_x)] + [V(r), x p_y] - [V(r), y p_x]$

$[q_\alpha, q_\beta] = 0$，$[p_\alpha, p_\beta] = 0$，$[q_\alpha, p_\beta] = \delta_{\alpha\beta}$，$[\theta, \varphi\psi] = \varphi[\theta, \psi] + \psi[\theta, \varphi]$

$= \frac{1}{2m}([p_x^2, x] p_y - [p_y^2, y] p_x) + [V(r), x] p_y + [V(r), p_y] x$

$- [V(r), y] p_x - [V(r), p_x] y$

$[V(r), x] = \sum_\alpha \frac{\partial V(r)}{\partial q_\alpha}\frac{\partial x}{\partial p_\alpha} - \frac{\partial V(r)}{\partial p_\alpha}\frac{\partial x}{\partial q_\alpha}$ $\because \frac{\partial V(r)}{\partial p_\alpha}, \frac{\partial x}{\partial p_\alpha} = 0 \Rightarrow [V(r), x] = 0$

$[V(r), p_y] = \frac{\partial V(r)}{\partial x}\frac{\partial p_y}{\partial p_x} + \frac{\partial V(r)}{\partial y}\frac{\partial p_y}{\partial p_y} + \frac{\partial V(r)}{\partial z}\frac{\partial p_y}{\partial p_z} = \frac{\partial V(r)}{\partial y}$

$= \frac{1}{2m}(2 p_x p_y - 2 p_y p_x) + x \cdot \frac{\partial V(r)}{\partial y} - y \frac{\partial V(r)}{\partial x}$

$= \frac{\partial V(r)}{\partial r}(x \cdot \frac{\partial r}{\partial y} - y \frac{\partial r}{\partial x})$

$= 2\frac{\partial V(r)}{\partial r}(x \cdot \frac{y}{r} - y \cdot \frac{x}{r}) = 0$

同理，$[H, L_y] = 0$，$[H, L_z] = 0$

以 L_z 为母函数的无限小变换为

$$\begin{cases} dx = \varepsilon \frac{\partial G}{\partial p_x} = -\varepsilon y \\ dy = \varepsilon \frac{\partial G}{\partial p_y} = \varepsilon x \\ dz = 0 \end{cases} \qquad \begin{cases} dp_x = -\varepsilon \frac{\partial G}{\partial x} = -\varepsilon p_y \\ dp_y = -\varepsilon \frac{\partial G}{\partial y} = \varepsilon p_x \\ dp_z = 0 \end{cases}$$

$$\begin{cases} X = x + dx = x - \varepsilon y \\ Y = y + \varepsilon x \\ Z = z \end{cases} \qquad \begin{cases} P_x = p_x + \varepsilon p_y \\ P_y = p_y - \varepsilon p_x \\ P_z = p_z \end{cases}$$

对应绕z轴的无限小转动

z

例：例 9.1，验证 $Q=\ln(\frac{1}{q}\sin p)$，$P=q\cot p$ 是正则变换，并求母函数 u_1

$$p\,dq-P\,dQ=p\,dq-q\cot p\,\frac{q}{\sin p}\cdot\frac{1}{q^2}(q\cos p\,dp-\sin p\,dq)$$
$$=p\,dq-q\cot^2 p\,dp+\cot p\,dq$$
$$=(p+\cot p)\,dq-q\cot^2 p\,dp$$
$$=\frac{\partial F}{\partial q}dq+\frac{\partial F}{\partial p}dp$$

$$F=pq+q\cot p+g(p)$$

$$\frac{\partial F}{\partial p}=q+q\cdot\frac{-1}{\sin^2 q}+g'(p)=-q\cot^2 p\ \Rightarrow\ g'(p)=0$$

$$\Rightarrow\ p\,dq-P\,dQ=d(pq+q\cot p)$$

$$u_1=pq+q\cot p=q\arcsin(e^Q q)+q\cdot\frac{\sqrt{1-q^2e^{2Q}}}{e^Q q}=q\arcsin(qe^Q)+\sqrt{q^2-e^{-2Q}}$$

9.7 验证：是正则变换，

$$\begin{cases}Q=\ln(1+\sqrt{q}\cos p)\\ P=2(1+\sqrt{q}\cos p)\sqrt{q}\sin p\end{cases}$$

并验证母函数 $u_2(p,Q)=-(e^Q-1)^2\tan p$

$$p\,dq-P\,dQ=p\,dq-2(1+\sqrt{q}\cos p)\sqrt{q}\sin p\,\frac{1}{1+\sqrt{q}\cos p}\left(\frac{1}{2\sqrt{q}}\cos p\,dq-\sqrt{q}\sin p\,dp\right)$$
$$=p\,dq-2\sqrt{q}\sin p\left(\frac{1}{2\sqrt{q}}\cos p\,dq-\sqrt{q}\sin p\,dp\right)$$
$$=p\,dq-\sin p\cos p\,dq+2q\sin^2 p\,dp$$
$$=(p-\sin p\cos p)\,dq+2q\sin^2 p\,dp$$
$$=\frac{\partial F}{\partial q}dq+\frac{\partial F}{\partial p}dp$$

$$F=\int(p-\sin p\cos p)\,dq+g(p)$$
$$=pq-\sin p\cos p\,q+g(p)$$

$$\frac{\partial F}{\partial p}=q-\cos 2p\cdot q+g'(p)=2q\sin^2 p\ \rightarrow\ g'(p)=0$$

$$F=pq-\sin p\cos p\,q$$

$S_3(p,Q) = -qp + F(q,Q) = -q\sin p\cos p = -(e^{Q}-1)\tan p$

$q = \left(\frac{e^{Q}-1}{\cos p}\right)^2$

9.3 $pdq - PdQ = pdq - q^{\alpha}\sin(\beta p)(\alpha q^{\alpha-1}\cos\beta p\, dq - q^{\alpha}\beta\sin\beta p\, dp)$

$= pdq - q^{2\alpha-1}\sin(\beta p)(\alpha\cos\beta p - \beta\sin\beta p)$

$\begin{vmatrix} \frac{\partial Q}{\partial q} & \frac{\partial Q}{\partial p} \\ \frac{\partial P}{\partial q} & \frac{\partial P}{\partial p} \end{vmatrix} = \alpha q^{\alpha-1}\cos\beta q\,\beta q^{\alpha}\cos\beta p + \alpha q^{\alpha-1}\sin(\beta p)\, q^{\alpha}\sin\beta p$

$= \alpha\beta q^{2\alpha-1} = 1 \qquad \alpha = \frac{1}{2},\ \beta = 2$

9.5 证明变换 $Q_\alpha = q_\alpha + m p_\alpha - e^{p_\alpha}$，$P_\alpha = p_\alpha$ 是正则变换，并求母函数 $U_3(q,P)$

$pdq - PdQ = pdq - p(dq + \frac{1}{p}dp - e^{p}dp)$

$= (-1 + pe^{p})dp$

$= \frac{\partial F}{\partial p}dq$

$F = \int(-1+pe^{p})dp = -p + (p-1)e^{p}$

$U_3(q,P) = PQ + F = (q + mp - e^{P})p - p + (p-1)e^{p}$

9.6. 证明变换 $Q_\alpha = q_\alpha$，$P_\alpha = p_\alpha + e^{-q_\alpha - 1}$ $\qquad q = -\ln(P-p) - 1$

$pdq - PdQ = pdq - (p + e^{-q-1})dq$

$= -e^{-q-1}dq = \frac{\partial F}{\partial q}dq$

$F = \int -e^{-q-1}dq = e^{-q-1}$

$U_4(p,P) = -qp + PQ + F = [\ln(P-p)+1](p+P) + (P-p)$

9.7 已知母函数 $U_1(q,Q) = -4q_1q_2 + 2e^{Q_1+2q_1} + 4e^{2Q_2+q_2}$，求正则变换

$p_1 = \frac{\partial U_1}{\partial q_1} = -4q_2 - 4e^{Q_1+2q_1}$, $\quad p_2 = \frac{\partial U_1}{\partial q_2} = -4q_1 + 4e^{2Q_2+q_2}$

$P_1 = -\frac{\partial U_1}{\partial Q_1} = -2e^{Q_1+2q_1}$, $\quad P_2 = -\frac{\partial U_1}{\partial Q_2} = -8e^{2Q_2+q_2}$

$Q_1 = \ln\left[\frac{1}{4}(P_1 + 4q_2)\right] - 2q_1$, $\quad Q_2 = \frac{1}{2}\ln\left[\frac{1}{4}(p_2 + 4q_1)\right] - q_2$

9.8. 已知母函数 $U_2(p,Q) = -p_1^2p_2^2 + \ln(Q_1+p_1) + \ln(Q_2+p_2)$

$q_1 = \frac{\partial U}{\partial p_1} = \cdots \frac{}{Q_1+p_1}$, $q_2 = \frac{\partial U}{\partial p_2} = \cdots \frac{}{Q_2+p_2}$

$P_1 = -\frac{\partial U_1}{\partial Q_1} = -\frac{1}{Q_1+p_1}$, $P_2 = -\frac{\partial U_1}{\partial Q_2} = -\frac{1}{Q_2+p_2}$

$Q_1 = \frac{1}{2p_1P_1^2 - q_1} - P_1$, $Q_2 = \frac{1}{2p_2^2P_2 - q_2} - P_2$

§9.2 哈密顿－雅可比方程

1. 哈密顿主函数

假设正则变换后所有正则变量都为常数（对应同到系统初始状态）

$K(P, Q, t) = 0$ 可满足这一点

$\Rightarrow \dot{P_\alpha} = -\frac{\partial K}{\partial Q_\alpha} = 0$, $\dot{Q_\alpha} = \frac{\partial K}{\partial P_\alpha} = 0$

$0 = K = H(q, p, t) + \frac{\partial U}{\partial t}$

若取 $U = U(q, P, t)$ 则 $p_\alpha = \frac{\partial U}{\partial q_\alpha}$

$\Rightarrow$ 哈－雅方程：

$s+1$ 个变量

$$H(\underline{q_1, q_2 \cdots q_s}; \frac{\partial U}{\partial q_1}, \frac{\partial U}{\partial q_2}, \cdots, \frac{\partial U}{\partial q_s}; \underline{t}) + \frac{\partial U}{\partial t} = 0$$

解（哈密顿主函数）：$S(q, P, t)$ 但方程中无 $\frac{\partial S}{\partial P}$

选取 P_1、$\cdots$、P_s 为 s 个积分常数，剩下 1 个加在 S 后面。

$$\frac{dS}{dt} = \sum_\alpha \underline{\frac{\partial S}{\partial q_\alpha}} \dot{q_\alpha} + \underline{\frac{\partial S}{\partial t}} \quad \to \quad S = \int L\, dt$$

$$= \sum_\alpha \underline{p_\alpha} \dot{q_\alpha} - \underline{H} = L$$

2. 哈密顿特征函数

设 H 不显含时间

令 $S(q, P, t) = W(q, P) + f(t)$

$\Downarrow$

$$H(q, \frac{\partial W}{\partial q}) = -f'(t) = E$$

哈－雅方程 分离变量 $\to$ $\begin{cases} f(t) = -Et \\ H(q, \frac{\partial W}{\partial q}) = E \end{cases}$

选择 W 为母函数时，

$K = H + \frac{\partial W}{\partial t} = H = E$

P : $E, C_2, \cdots C_s$

Q : 均为可遗坐标

3. 可分离系统

若 H 可写为 $H=(q,p) = H(q_\alpha, \frac{\partial W}{\partial q_\alpha}, \phi(q_1, \frac{\partial W}{\partial q_1}))$

哈-雅方程：$H(q_\alpha, \frac{\partial W}{\partial q_\alpha}, \phi(q_1, \frac{dW}{dt})) - E = 0$

设 $W = W'(q_\alpha) + W_1(q_1)$

则 $H(q_\alpha, \frac{\partial W'}{\partial q_\alpha}, \phi_1(q_1, \frac{dW_1}{dt})) - E = 0$

$$\rightarrow \begin{cases} \phi_1(q_1, \frac{dW_1}{dt}) = C_2 \\ H(q_\alpha, \frac{\partial W'}{\partial q_\alpha}, C_2) - E = 0 \end{cases}$$

(1) 对于完全可分离系统（每个变量都可以分离）

特征函数 $W = \sum_\alpha W_\alpha(q_\alpha, E, C_2, \cdots, C_s)$

哈-雅方程：
$$\begin{cases} \phi_1(q_1, \frac{dW_1}{dq_1}, E, C_2, \cdots, C_s) = 0 \\ \vdots \\ \phi_s(q_s, \frac{dW_s}{dq_s}, E, C_2, \cdots, C_s) = 0 \end{cases}$$

(2) 当 q_α 是可遗坐标时，$W_\alpha = C q_\alpha$

证：此时 $p_\alpha = \frac{\partial W_\alpha}{\partial q_\alpha} = C$ $\quad \frac{d}{dt}(p_\alpha) = 0$

(3) 若系统哈密顿量是"可分离的"

$$H(q; p; t) = \sum_\alpha H_\alpha(q_\alpha, p_\alpha)$$

例：用哈-雅方程求解谐振子问题

$$H = \frac{1}{2m} p^2 + \frac{1}{2} k x^2$$

① H 不显含时间 → 特征函数

哈-雅方程 $H = \frac{1}{2m}\left(\frac{\partial W}{\partial x}\right)^2 + \frac{1}{2} k x^2 = E$

$$\rightarrow \frac{dW}{dx} = \sqrt{2mE - mkx^2} = \sqrt{mk}\sqrt{\frac{2E}{k} - x^2}$$

$$\rightarrow W = \sqrt{mk}\int\sqrt{\frac{2E}{k} - x^2}\, dx$$

求各个动量 (E, C_2) 对应坐标

哈密顿正则方程：$\dot{X} = \frac{\partial K}{\partial E} = 1 \rightarrow X = t - t_0$

正则变换：$X = \frac{\partial W}{\partial E}$

$$= \sqrt{mk}\int \frac{1}{2}\cdot\frac{\frac{2}{k}}{\sqrt{\frac{2E}{k} - x^2}}\, dx \qquad \int\frac{dx}{\sqrt{a - x^2}} \quad x = \sqrt{a}\sin t$$

$$= \sqrt{\frac{m}{k}}\int\frac{dx}{\sqrt{\frac{2E}{k} - x^2}} \qquad = \int\frac{\sqrt{a}\cos t}{\sqrt{a}\cos t}\, dt = \arcsin\left(\frac{x}{\sqrt{a}}\right)$$

$$= \sqrt{\frac{m}{k}}\arcsin\left(\sqrt{\frac{k}{2E}}\, x\right)$$

$$\rightarrow x = \sqrt{\frac{2E}{k}}\sin\left(\sqrt{\frac{k}{m}}\, X\right) = \sqrt{\frac{2E}{k}}\sin\sqrt{\frac{k}{m}}(t - t_0)$$

② 主函数

$$\frac{1}{2m}\left(\frac{\partial S}{\partial x}\right)^2 + \frac{1}{2} kx^2 + \frac{\partial S}{\partial t} = 0$$

$$S = W - Et = \sqrt{mk}\int\sqrt{\frac{2E}{k} - x^2}\, dx - Et$$

$$K = H + \frac{\partial S}{\partial t} = H - H = 0$$

$$\dot{X} = \frac{\partial K}{\partial E} = 0 \rightarrow \dot{X} = C_2$$

$$X = \frac{\partial S}{\partial E} = \sqrt{\frac{m}{k}}\int\frac{dx}{\sqrt{\frac{2E}{k} - x^2}} - t$$ （回到原来的正则变量）

$$= \sqrt{\frac{m}{k}}\arcsin\left(\sqrt{\frac{k}{2E}}\, x\right) - t$$

$\Rightarrow x = \frac{2E}{k}\sin\left[\sqrt{\frac{k}{m}}(\alpha + C_2)\right]$

例：用哈密顿-雅可比方程研究平方反比的有心吸引力问题

$H = \frac{1}{2m}\left(p_\rho^2 + \frac{1}{\rho^2}p_\varphi^2\right) - \frac{mk^2}{\rho}$

H不显含时间 → 哈密顿特征函数

$$H = \frac{1}{2m}\left[\left(\frac{\partial W}{\partial \rho}\right)^2 + \frac{1}{\rho^2}\left(\frac{\partial W}{\partial \varphi}\right)^2\right] - \frac{mk^2}{\rho} = E$$

将W分离：$W(\rho,\varphi,E,C_2) = W_1(\rho,E,C_2) + W_2(\varphi,E,C_2)$

$$\hookrightarrow \frac{1}{2m}\left[\left(\frac{\partial W_1}{\partial \rho}\right)^2 + \frac{1}{\rho^2}\left(\frac{\partial W_2}{\partial \varphi}\right)^2\right] - \frac{mk^2}{\rho} - E = 0$$

$$\hookrightarrow \underbrace{\left(\frac{\partial W_2}{\partial \varphi}\right)^2(\varphi, E, C_2)}_{\varphi\text{的函数}} = \underbrace{-\rho^2\left(\frac{\partial W_1}{\partial \rho}\right)^2 + 2m^2k^2\rho + 2mE\rho^2}_{\rho\text{的函数}} = 0$$

$$\Rightarrow \begin{cases} \frac{\partial W_2}{\partial \varphi} = C_2 & ① \\ -\rho^2\left(\frac{\partial W_1}{\partial \rho}\right)^2 + 2m^2k^2\rho + 2mE\rho^2 = C_2^2 & ② \end{cases}$$

① → $W_2 = C_2\varphi$ （φ是可遗坐标，C_2为对应动量）

② → $\frac{dW_1}{d\rho} = \sqrt{2mE + \frac{2m^2k^2}{\rho} - \frac{C_2^2}{\rho^2}}$

$W_1 = \int\sqrt{2mE + \frac{2m^2k^2}{\rho} - \frac{C_2^2}{\rho^2}}\,d\rho$

$W = C_2\varphi + \int\sqrt{2mE + \frac{2m^2k^2}{\rho} - \frac{C_2^2}{\rho^2}}\,d\rho$ 为正则变换的母函数

$K = H + \frac{\partial W}{\partial t} = H$

$\dot{Q}_1 = \frac{\partial K}{\partial E} = 1$, $\dot{Q}_2 = \frac{\partial K}{\partial C_2} = 0$

$\hookrightarrow Q_1 = t + C_1' = \frac{\partial W}{\partial E}$ $\quad Q_2 = C_2' = \frac{\partial W}{\partial C_2}$

$$\rightarrow \begin{cases} \int \frac{m\,d\rho}{\sqrt{2mE + \frac{2m^2k^2}{\rho} - \frac{C_2^2}{\rho^2}}} = t + C_1' \\ \varphi - \int \frac{C_2\,d\rho}{\rho^2\sqrt{2mE + \frac{2m^2k^2}{\rho} - \frac{C_2^2}{\rho^2}}} = C_2' \end{cases}$$

§9.3 作用量变量与角变量

讨论每个自由度都做周期运动且完全可分离的系统

① 天平动：p、q 都是时间周期函数且周期相同

② 转动：p 是 q 的周期函数（q 每增加 q_0，系统状态重现）

q 一般为角度

定义：作用量 J

$$J_\alpha = \oint p_\alpha dq_\alpha = \oint \frac{\partial W_\alpha(q_\alpha, E, C_2 \cdots C_s)}{\partial q_\alpha} dq_\alpha$$

$$\hookrightarrow W = \sum_\alpha W_\alpha(q_\alpha; J_1 \cdots J_s)$$

定义：角变量 w

以 W 为正则变换的母函数，J_α 为"动量"

对应坐标 $w_\alpha = \frac{\partial W}{\partial J_\alpha}$

变换后哈密顿量 $K = H + \frac{\partial W}{\partial t} = H = E$

变换后的哈密顿方程：

$$\dot{w}_\alpha = \frac{\partial K}{\partial J_\alpha} = \frac{\partial E}{\partial J_\alpha}$$

$$\hookrightarrow w_\alpha = \left(\frac{\partial E}{\partial J_\alpha}\right)t + w_{\alpha 0}$$

考虑一个周期：

$$\Delta w_\alpha = \oint dw_\alpha = \oint \frac{\partial w_\alpha}{\partial q_\alpha} dq_\alpha = \oint \frac{\partial^2 W}{\partial J_\alpha \partial q_\alpha} dq_\alpha$$

$$= \oint \frac{\partial p_\alpha}{\partial J_\alpha} dq_\alpha$$

$$= \frac{\partial}{\partial J_\alpha} \oint p_\alpha dq_\alpha = 1$$

$\Rightarrow \frac{\partial E}{\partial J_\alpha} \cdot T_\alpha = 1$ 　频率 $\nu_\alpha = \frac{\partial E}{\partial J_\alpha}$

例：求谐振子的频率 $H = \frac{1}{2m}p^2 + \frac{1}{2}kx^2 = E$

$$J_\alpha = \oint p\,dx$$

$$= \oint \sqrt{2mE - mkx^2}\,dx$$

$$= 2\sqrt{mk}\int_{-\sqrt{\frac{2E}{k}}}^{\sqrt{\frac{2E}{k}}}\sqrt{\frac{2E}{k} - x^2}\,dx \qquad x = \sqrt{\frac{2E}{k}}\sin t$$

$$= 2\sqrt{mk}\cdot\frac{2E}{k}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos^2 t\,dt$$

$$= \frac{4E}{k}\sqrt{mk}\int_0^{\frac{\pi}{2}}[1+\cos(2t)]\,dt$$

$$= \frac{4E}{k}\sqrt{mk}\,\frac{\pi}{2} = 2\pi E\sqrt{\frac{m}{k}}$$

$$\nu_\alpha = \frac{\partial E}{\partial J_\alpha} = \frac{1}{2\pi}\sqrt{\frac{k}{m}} = f_\alpha$$

$$T_\alpha = \frac{1}{f_\alpha} = 2\pi\sqrt{\frac{m}{k}}$$

例：求平方反比的有心吸引力作用下质点的运动频率

$$H = \frac{1}{2m}\left(p_\rho^2 + \frac{1}{\rho^2} p_\varphi^2\right) - \frac{mk^2}{\rho}$$

作用量 $J_\varphi = \oint p_\varphi \, d\varphi =$

§ 9.φ

补充：辛几何

构造正则变量 $\eta = \begin{pmatrix} q \\ p \end{pmatrix}$

Hamilton 方程：$\dot{\eta} = \begin{pmatrix} \frac{\partial H}{\partial p} \\ -\frac{\partial H}{\partial q} \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & I_s \\ -I_s & 0 \end{pmatrix}}_{J} \begin{pmatrix} \frac{\partial H}{\partial p} \\ \frac{\partial H}{\partial q} \end{pmatrix} = J \frac{\partial H}{\partial \eta}$

对于正则变换 $\eta = \begin{pmatrix} q \\ p \end{pmatrix} \to \xi = \begin{pmatrix} Q \\ P \end{pmatrix}$，仍有正则方程 $\dot{\xi} = J \frac{\partial H}{\partial \xi}$

记为 $M = \frac{\partial \xi}{\partial \eta}$ $\quad M_{ij} = \frac{\partial \xi_i}{\partial \eta_j}$

$$\dot{\xi} = \frac{\partial \xi}{\partial \eta} \dot{\eta} = M \dot{\eta}$$

$$\frac{\partial H}{\partial \eta_i} = \frac{\partial H}{\partial \xi_j} \frac{\partial \xi_j}{\partial \eta_i} = M^T_{ij} \frac{\partial H}{\partial \xi_j} \to \frac{\partial H}{\partial \eta} = M^T \frac{\partial H}{\partial \xi}$$

$$\begin{cases} \dot{\eta} = J \frac{\partial H}{\partial \eta} \\ \dot{\xi} = J \frac{\partial H}{\partial \xi} \end{cases} \Rightarrow \quad M^{-1} J \frac{\partial H}{\partial \xi} = J M^T \frac{\partial H}{\partial \xi}$$

$J = MJM^T$ 正则变换是保辛结构

辛矩阵

$$[u, v]_{pq} = \left(\frac{\partial u}{\partial q} \ \frac{\partial u}{\partial p} \right) \begin{pmatrix} & I_s \\ -I_s & \end{pmatrix} \begin{pmatrix} \frac{\partial v}{\partial q} \\ \frac{\partial v}{\partial p} \end{pmatrix}$$

$$= \left(\frac{\partial u}{\partial \eta}\right)^T J \left(\frac{\partial v}{\partial \eta}\right) \qquad \frac{\partial}{\partial \eta} = M^T \frac{\partial}{\partial \xi}$$

$$= \left(M^T \frac{\partial u}{\partial \xi}\right)^T J \left(M^T \frac{\partial v}{\partial \xi}\right)$$

$$= \left(\frac{\partial u}{\partial \xi}\right)^T M J M^T \frac{\partial v}{\partial \xi}$$

$= (\delta\xi)\ J\ \delta\xi = [u, v]\, PQ$ [illegible]